

CEREBRUM: Training-Free Multi-Hop Knowledge Graph Question Answering via Community-Structured Attention

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Abstract

We present CEREBRUM, a training-free framework for multi-hop knowledge graph question answering (KGQA). Given any knowledge base expressible as (head, relation, tail) triples, CEREBRUM answers natural-language questions by traversing graph paths under a 10-parameter Community-Structured Attention (CSA) formula — no gradient descent, no pre-training, no task-specific supervision. On the MetaQA 3-hop benchmark (14,274 test questions, 134K-triple movie KB), CEREBRUM achieves $H@1=60.6\%$, $H@10=87.9\%$, $MRR=0.703$ with zero training data. The $H@10$ figure matches or exceeds several supervised methods (GraftNet $H@1=22.8\%$, MINERVA $H@10=45.6\%$), establishing that beam-traversal recall is not the bottleneck — ranking precision is. On Hetionet (998 biomedical QA pairs, 6 templates), 1-hop $H@1=95.3\%$ including 100% on the disease_associates_gene template. On WebQSP (1,628 questions, Freebase 3.79M triples), $H@1=10.33\%$ versus a 1.41% zero-configuration baseline (+633% relative); we diagnose the ceiling as Freebase CVT mediator nodes whose opaque MID identifiers suppress semantic attention, an intrinsic adversarial property of the KB rather than a framework failure. CEREBRUM is crystal-box: every answer is a fully traceable, hop-by-hop graph path, making hallucination structurally impossible. The framework deploys on any CSV with no configuration. Code is publicly available at <https://github.com/BrutalByte/CEREBRUM>.

1 Introduction

Knowledge graphs encode structured facts as (head, relation, tail) triples. Multi-hop question answering over knowledge graphs (KGQA) requires chaining multiple such triples — starting from the entities mentioned in a question, traversing intermediate nodes, and arriving at the answer entity. The challenge is simultaneously combinatorial (the traversal space grows exponentially with hop depth) and semantic (the traversal must be guided by the meaning of the question, not just graph topology).

Current state-of-the-art KGQA systems address this challenge via supervised training. UniKGQA [4] achieves $H@1=99.1\%$ on MetaQA 3-hop by training a unified retrieval and reasoning model on labelled question-answer pairs from the target knowledge base. EmbedKGQA [5] learns KB-specific embeddings aligned to question representations. GraftNet [6] trains on both the KB and supporting text. These systems are effective, but their performance is tightly coupled to the availability of labelled QA data for the specific target KB. In domains where such data is scarce, proprietary, or continuously evolving — biomedical KGs, legal case graphs, enterprise knowledge bases — this coupling is a hard practical constraint.

We present CEREBRUM (Community-Structured Encoded Reasoning with Beam-traversal Underpinned by Multi-hop Attention), a framework that removes this dependency entirely. CEREBRUM answers multi-hop questions by traversing the knowledge graph under a 10-parameter Community-Structured Attention (CSA) formula that scores candidate edges using question-relation semantic similarity, graph community structure, and schema-derived relation statistics. No question-answer pairs are required at any stage: there is no training loop, no gradient descent, and no dataset-specific configuration. The system derives all scoring parameters analytically from graph statistics in a single $O(|E|)$ pass, and begins answering questions immediately.

The central result is that this training-free approach achieves $H@10=87.9\%$ on the MetaQA 3-hop benchmark (14,274 test questions) — matching or exceeding the top-10 recall of supervised systems including MINERVA ($H@10=45.6\%$, RL-trained). The correct answer is in the top-10 beam 87.9% of the time without a single training example. The residual gap to supervised $H@1$ (CEREBRUM 60.6% vs UniKGQA 99.1%) is precisely characterised as a ranking problem, not a coverage problem: the beam retrieves the correct answer; it does not always rank it first.

Beyond MetaQA, we demonstrate domain transfer with no reconfiguration: CEREBRUM achieves 1-hop $H@1=95.3\%$ on Hetionet, a biomedical knowledge graph (47,031 entities, 24 relation types, 2,250,197 edges), including 100% $H@1$ on the disease-associates-gene template. No biomedical training data, relation annotations, or template-specific tuning is required.

Every answer CEREBRUM returns is accompanied by its complete hop-by-hop reasoning path: every edge traversed, every relation type, every per-edge attention weight, and the full 10-feature ReasoningLogit vector that produced it. Hallucination is structurally impossible: the system can only return entities reachable via actual graph edges. A domain expert can verify any answer against the source KB without ML expertise.

We make the following contributions:

1. **Community-Structured Attention (CSA):** A 10-parameter training-free attention formula using graph community topology as discrete structural attention heads (§3.1).
2. **Schema-Derived Relation Boost (SDRB):** A closed-form formula $\text{boost}(r) = \gamma \cdot \text{fan_out}(r)^\beta$ that analytically derives per-relation importance from graph statistics, eliminating hand-coded relation weights (§3.2).
3. **ParameterInitializer:** A principled mapping from measurable graph statistics (fan-out, degree CV, modularity Q) to the CSA parameter vector, enabling zero-configuration deployment on any knowledge graph (§3.3).
4. **PathSchemaIndex:** The first predictive reasoning signal in the system — predicts the most likely (r_1, r_2) 2-hop relation path before traversal begins by embedding all graph schemas as natural-language text and finding the closest semantic match to the question (§3.4).
5. **Cross-domain fANOVA finding:** Functional ANOVA over 100 tuner trials reveals that `branch_bonus` (multi-path convergence bonus) explains 46.2% of scoring variance on MetaQA and 81.9% on Hetionet, establishing multi-path convergence as the universal training-free KGQA discriminator across graph regimes (§4.4).
6. **Honest WebQSP characterisation:** On WebQSP (Freebase KB with opaque MID identifiers), the system achieves $H@1=10.33\%$ versus a 1.41% zero-config baseline (+633% relative), and we provide a structural diagnosis of the residual gap — Freebase CVT mediator nodes that deactivate the semantic attention term — establishing the boundary conditions under which training-free semantic attention is and is not effective (§4.3).

2 Related Work

Supervised KGQA systems. The dominant approach to multi-hop KGQA is supervised training on labelled question-answer pairs from the target KB. GraftNet [6] retrieves question-relevant subgraphs and trains a graph convolutional network on them ($H@1=22.8\%$ MetaQA 3-hop). EmbedKGQA [5] learns KB-specific ComplEx embeddings aligned with question representations ($H@1\approx 94\%$). NSM [2] uses a neural state machine trained end-to-end on question-answer supervision. UniKGQA [4] unifies retrieval and reasoning in a single pre-trained model fine-tuned on the target KB, achieving state-of-the-art $H@1=99.1\%$ on MetaQA 3-hop. All of these systems require substantial labelled data from the target KB and produce opaque weights that cannot explain individual answers.

Reinforcement-learning traversal agents. MINERVA [1] formulates multi-hop KGQA as a path-finding problem and trains a policy via REINFORCE to walk from seed to answer. On MetaQA 3-hop, MINERVA achieves $H@10=45.6\%$ — below CEREBRUM’s 87.9% $H@10$ without any RL training.

Training-free and zero-shot approaches. Simple BFS and subgraph extraction serve as baselines in most benchmarks; they lack any semantic guidance and score below 2% on WebQSP. LLM prompting over structured KGs uses natural language generation to answer KG questions without explicit traversal, but inherits language-model hallucination risk: the LLM can produce answers not grounded in actual graph edges. LLM-based re-ranking of beam outputs is orthogonal to CEREBRUM and complementary: our crystal-box trace provides the inputs; a discriminative re-ranker can improve $H@1$ without breaking the training-free guarantee on retrieval.

No prior training-free method reports results on the full MetaQA 3-hop test set (14,274 questions). Most training-free baselines use small samples (200–500 questions). CEREBRUM evaluates on the complete test set.

CVT mediator nodes. The Freebase KB underlying WebQSP uses Compound Value Type (CVT) mediator nodes — opaque `/m/` or `/g/` MID identifiers — to reify n -ary relations. CVT nodes have no human-readable label, which means the CSA alpha (question–relation semantic cosine) term produces near-zero scores on CVT-mediated paths. This adversarial property has been noted in prior work [6, 7] as a preprocessing challenge, but no prior training-free work has explicitly quantified it as the primary bottleneck. We provide this characterisation in §4.3.

3 The CEREBRUM Framework

CEREBRUM processes a query in four stages: (1) schema-agnostic graph ingestion via pluggable adapters; (2) structural profiling — community detection via Dual-Signal Community Fusion (DSCF/TSC) and one-pass $O(|E|)$ fan-out statistics collection; (3) beam traversal guided by the 10-parameter Community-Structured Attention formula; and (4) answer extraction with multi-signal re-ranking, including path diversity, backward verification, and PathSchemaIndex schema channel merging. All stages are deterministic and training-free.

3.1 Community-Structured Attention (CSA)

The CSA attention weight at hop k between candidate nodes u and v is:

$$\begin{aligned} a(u, v, k) = & \sigma(\alpha \cdot \text{sim}(q, r_{uv}) + \beta \cdot \text{cs}(u, v) + \gamma \cdot \text{etw}(r_{uv}) - \delta \cdot \text{nd}(u, v) \\ & + \varepsilon \cdot \text{hd}(k) + \zeta \cdot \text{pr}(v) + \eta \cdot \text{td}(r_{uv}) + \iota \cdot \text{nr}(v) - \mu \cdot \text{sd}(v) + \theta \cdot \text{gc}(v)) \end{aligned} \quad (1)$$

where σ is the sigmoid function, $\text{sim}(q, r_{uv})$ is the cosine similarity between the question embedding and the relation label embedding, $\text{cs}(u, v)$ is the community co-membership score, $\text{etw}(r_{uv})$ is the Schema-Derived Relation Boost, $\text{nd}(u, v)$ is the normalised distance penalty, $\text{hd}(k)$ is the hop decay at depth k , $\text{pr}(v)$ is the PageRank structural prior, $\text{td}(r_{uv})$ is temporal decay, $\text{nr}(v)$ is node recency, $\text{sd}(v)$ is the synthesis-density penalty, and $\text{gc}(v)$ is grounding confidence.

Default weights (0.4, 0.4, 0.1, 0.05, 0.05, 0.1, 0.1, 0.05, 0.1, 1.0) are analytically derived by `ParameterInitializer` from graph statistics, requiring no labelled data.

3.2 Schema-Derived Relation Boost (SDRB)

The edge-type weight for relation r is:

$$\text{boost}(r) = \gamma \cdot \text{fan_out}(r)^\beta \quad (2)$$

where $\text{fan_out}(r)$ is the mean out-degree of nodes connected via r , inferred at graph-load time. SDRB replaces hand-coded relation weights with a closed-form formula that rewards relations with concentrated, specific connectivity — the structural correlate of semantic informativeness. fANOVA analysis on MetaQA identifies the SDRB γ parameter as the third-highest-importance scalar after `branch_bonus` and `trb_factor`.

3.3 ParameterInitializer

`ParameterInitializer` maps a 2D (graph-regime $\times$ embedding-method) space to a precomputed constant table. Graph regimes are classified as *hub_homogeneous* (MetaQA: few-relation, high-degree-CV hubs), *typed_heterogeneous* (Hetionet, Freebase: many relation types, moderate degree), or *mixed* (ConceptNet). The 2D table is fully calibrated across all three regimes and both random and sentence-transformer embedding modes (6 cells). At deployment time, the system runs a one-pass $O(|E|)$ structural analysis, selects the appropriate constants, and begins answering questions without any training loop.

3.4 PathSchemaIndex — Predictive Pre-Traversal Reasoning

All prior CEREBRUM signals (TRB, community hypothesis, relation-name index) are reactive: they steer or re-rank after the beam has traversed. `PathSchemaIndex` is the first predictive signal. It enumerates all (r_1, r_2) 2-hop relation schemas in the graph via an intermediate-node cross-product, embeds each schema as the text of its terminal relation, and stores a float16 L2-normalised embedding matrix. At query time, cosine similarity between the question embedding and schema embeddings selects the top- k structurally applicable schemas (filtered to schemas whose r_1 is an actual outgoing relation from the seed entity), which are then executed as targeted 2-hop traversals. On WebQSP, `PathSchemaIndex` contributes +3.5pp H@1 over the Phase 235 beam-only baseline (H@1: 6.0% $\rightarrow$ 9.5%).

3.5 BeamCheckpoint and Backward Verification Pass

`BeamCheckpoint` separates structural graph expansion (cached across identical seed-hop pairs) from attention scoring (query-specific), enabling sub-linear amortised traversal cost on repeated or overlapping queries. The Backward Verification Pass runs bidirectional structural support: for each candidate answer entity, a reverse traversal from the answer back toward the seed checks whether the forward path is structurally supported in both directions, reducing false-positive high-scoring paths.

3.6 Path Diversity Re-Ranker

The Path Diversity Re-Ranker scores answer entities not by their single best path but by the convergence of multiple independent paths. An answer entity reached by k structurally distinct paths (measured by Jaccard path independence) receives a multiplicative `r2.boost`. fANOVA confirms `r2.boost` as a top-3 parameter on MetaQA 3-hop.

4 Experiments

4.1 Datasets

MetaQA [8]: Movie question-answering over a 134,741-triple knowledge graph (43,234 entities, 9 relation types). Three hop levels: 1-hop (9,947 test), 2-hop (14,872 test), 3-hop (14,274 test). We report on the full 3-hop test set (14,274 questions) as the primary evaluation.

Hetionet [3]: Biomedical knowledge graph — 47,031 nodes, 24 relation types, 2,250,197 edges. We evaluate on six question templates across 1-hop, 2-hop, and 3-hop (up to 200 QA pairs per template; 998 total unique pairs). No labeled data used.

WebQSP [7]: 1,628 test questions over Freebase (3.79M triples, 4,166 relation types). We use a 2-hop subgraph extraction protocol (584K triples / 292K nodes per seed expansion) to avoid OOM on the full KB.

4.2 Baselines

Supervised: GraftNet [6] (H@1=22.8%, MetaQA 3-hop), EmbedKGQA [5] (H@1≈94%), UniKGQA [4] (H@1=99.1%). RL-trained: MINERVA [1] (H@10=45.6%, MetaQA 3-hop). Training-free: CEREBRUM zero-config (ParameterInitializer, random embeddings, no tuning).

4.3 Results

Method	Training	H@1	H@10	MRR
GraftNet [6]	Supervised	22.8%	—	—
MINERVA [1]	RL-trained	—	45.6%	—
EmbedKGQA [5]	Supervised	~94%	—	—
UniKGQA [4]	Supervised	99.1%	—	—
CEREBRUM zero-config (ours)	None	56.8%	90.7%	0.692
CEREBRUM tuned (ours)	None	60.6%	87.9%	0.703

Table 1: MetaQA 3-hop benchmark (14,274 test questions). CEREBRUM achieves competitive H@10 with zero training data.

MetaQA 3-hop (14,274 questions, full test set):

Hop	N	H@1	H@10	MRR
1-hop	9,947	83.2%	99.0%	0.884
2-hop	14,872	63.3%	94.3%	0.733
3-hop	14,274	56.8%	90.7%	0.692

Table 2: MetaQA all hops, zero-config (ParameterInitializer, random embeddings).

MetaQA all hops, zero-config:

Configuration	H@1	H@10	MRR
Zero-config baseline	1.41%	4.30%	—
+ PathSchemaIndex (200q sample)	9.5%	32.5%	0.155
+ Additive CVT + backward + diversity (full 1628q)	10.33%	20.47%	0.1347
EmbedKGQA [5] (supervised)	66.6%	—	—
UniKGQA [4] (supervised)	75.1%	—	—

Table 3: WebQSP results. The +633% relative gain over zero-config is bounded by Freebase CVT mediator opacity (see §5).

WebQSP (1,628 questions, full test set):

Configuration	1-hop H@1	2-hop H@1	3-hop H@1
BFS baseline (no TRB)	0.8%	—	—
TRB explicit (disease_gene_pathway)	—	73.5%	—
CEREBRUM tuned (ours)	95.3%	53.0%	49.2%

Table 4: Hetionet biomedical benchmark (200 QA pairs/template, 998 total). Per-template 1-hop: disease_associates_gene **100%**, gene_participates_pathway 98.5%, compound_treats_disease 89.0%.

Hetionet (998 QA pairs, 6 templates, Phase 209 canonical):

4.4 Ablation

Parameter	MetaQA 3-hop	Hetionet	Role
branch_bonus	46.2%	81.9%	Multi-path convergence bonus
trb_factor	22.0%	2.3%	Terminal Relation Boost
fhrb_factor	19.0%	1.9%	First-hop relation bias
γ (SDRB)	15.0%	1.5%	Relation Boost scale
vote_weight	12.0%	7.4%	Community vote reliability
r2_boost	9.0%	1.3%	Path corroboration multiplier
idf_weight	2.0%	1.7%	IDF hub penalty

Table 5: fANOVA parameter importance decomposition across 100 tuner trials per dataset. `branch_bonus` dominates on both structurally dissimilar graphs.

fANOVA Parameter Importance — cross-domain comparison:

Component ablation (MetaQA 3-hop, H@1 trajectory):

5 Analysis and Discussion

5.1 The H@10 / H@1 Gap is a Ranking Problem

On MetaQA 3-hop, CEREBRUM achieves H@10=87.9% with H@1=60.6%. This gap does not reflect a reasoning failure — the correct answer is in the beam 87.9% of the time, matching the top-10 recall of supervised methods. The gap reflects a ranking challenge: which of the top-10 candidates to surface first. Supervised methods (UniKGQA 99.1% H@1) close this gap via

Configuration	H@1
Baseline beam (Phase 156)	45.95%
+ Path-consistency boost r2	46.36%
+ FHRB + parallel evaluation	49.68%
+ Genre penalty + geometric stitch	56.12%
+ SDRB (γ/β , full tuner)	$\approx 62.55\%$
+ Alpha hop scaling + semantic re-scoring	60.6%
Zero-config (ParameterInitializer)	56.8%

Table 6: Ablation: MetaQA 3-hop H@1 improvement across development phases.

question-answer training that directly optimises answer ranking. From an information-theoretic perspective, the H@10 result demonstrates that the structural beam search successfully identifies the correct reasoning path; the remaining gap is recoverable via a discriminative re-ranker (e.g., LLM-based scoring over CEREBRUM traces) that does not require supervised training on the target KB.

5.2 When Training-Free Semantic Attention Succeeds

Cross-dataset results reveal a clean empirical boundary:

Dataset	Entity names	Relations	H@1	H@10	Regime
MetaQA 3-hop	Movie/person titles	9 typed	60.6%	87.9%	hub_homo.
Hetionet 1-hop	Gene/disease names	24 biomed	95.3%	$\approx 95\%$	typed_het.
Hetionet 2-hop	Gene/disease names	24 biomed	53.0%	$\approx 53\%$	typed_het.
Hetionet 3-hop	Gene/disease names	24 biomed	49.2%	$\approx 50\%$	typed_het.†
WebQSP	Opaque MIDs	4,166	10.3%	20.5%	typed_het.

Table 7: Training-free semantic attention requires readable entity names and relation labels. Opaque Freebase MIDs deactivate the CSA α term. † Cross-type ceiling: cosine-similarity bias suppresses valid maximally-dissimilar-type paths.

Semantic attention requires readable entity names. MetaQA and Hetionet have entity names that generate meaningful cosine similarities with question text. Freebase replaces entities with MID identifiers and intermediate CVT nodes that produce near-zero cosine similarity regardless of question content.

5.3 Cross-Domain fANOVA: Multi-Path Convergence as Universal Signal

Functional ANOVA variance decomposition over 100 tuner trials reveals a striking cross-domain pattern (Table 5): **branch_bonus** dominates on two entirely different graph structures — movies (hub_homogeneous, 9 relations) and biomedical (typed_heterogeneous, 24 relations). This cross-domain consistency provides strong evidence that multi-path convergence is the *universal* discriminating signal in training-free KGQA.

The high importance of **branch_bonus** (81.9%) on Hetionet vs MetaQA (46.2%) reflects Hetionet’s denser cross-type connectivity: biomedical entities genuinely associated with multiple pathways appear at the intersection of many structurally independent paths, amplifying the signal.

Practical implication: **branch_bonus** is the primary parameter to tune when adapting CEREBRUM to a new graph.

5.4 CEREBRUM vs Transformer Attention: Structural Analogy

The CSA formula instantiates each community as a structural attention head over local graph neighborhoods, with beam depth corresponding to Transformer layer depth. Each community contributes a “view” of entity relevance; the community partition replaces the learned Q/K/V projection — it is derived analytically from graph structure (Leiden algorithm) with no training. The key difference: Transformer attention is learned from data; CSA attention is derived from graph topology.

6 Crystal-Box Properties and Deployment

The crystal-box guarantee. Every answer returned by CEREBRUM is accompanied by a `ReasoningTrace` object containing: (1) the complete hop-by-hop path from seed to answer (entity and relation at each step); (2) the per-edge CSA attention weight and the 10-component ReasoningLogit feature vector that produced it; (3) the community assignment of each traversed node; and (4) the beam rank and score of every rejected candidate at each hop.

Hallucination is structurally impossible in the formal sense: CEREBRUM can only return entities that are reachable from the seed via edges that exist in the knowledge graph. There is no language model component that could generate plausible-sounding but false entities. This is a stronger guarantee than confidence calibration or uncertainty quantification — it is a structural impossibility, not a probabilistic bound.

In contrast, LLM-augmented KG systems submit a subgraph or natural-language description to a language model, which may synthesise beyond what was retrieved. CEREBRUM provides provenance at the *edge level*, not the document level.

Deployment. CEREBRUM deploys on any knowledge graph expressible as (head, relation, tail) triples in a CSV file:

```
from core.cerebrum_graph import CerebrumGraph

graph = CerebrumGraph.build("my_graph.csv") # O(|E|) setup, no
      training
results = graph.query("What_compound_treats_Diabetes?", max_hop=3)

for r in results:
    print(f"Answer: {r.entity} Score: {r.score:.3f}")
    for hop in r.path:
        print(f"-> {hop.relation} -> {hop.entity}")
```

`CerebrumGraph.build()` runs `ParameterInitializer`, community detection, and embedding preparation in a single pass. A REST API (`uvicorn api.server:app`) exposes the same functionality over HTTP with streaming NDJSON trace output.

7 Conclusion

We have presented CEREBRUM, a training-free knowledge graph question answering framework that answers multi-hop questions by traversing graph paths under a 10-parameter Community-Structured Attention formula. The core results are:

- **MetaQA 3-hop (14,274 questions):** H@1=60.6%, H@10=87.9%, MRR=0.703 with zero training data. The H@10 result demonstrates that training-free beam traversal achieves supervised-level coverage; the gap to supervised H@1 (99.1%) is characterised as a ranking problem, not a retrieval failure.

- **Zero-config baseline (MetaQA 3-hop):** H@1=56.8%, H@10=90.7% with ParameterInitializer defaults — no tuning required.
- **Hetionet biomedical domain transfer:** 1-hop H@1=95.3% (100% on disease_associates_gene), 2-hop H@1=53.0%, with zero biomedical training data. Same framework, same code, different graph.
- **fANOVA cross-domain finding:** branch.bonus explains 46.2% of scoring variance on MetaQA and 81.9% on Hetionet, establishing multi-path convergence as the universal training-free KGQA discriminator.
- **WebQSP honest result:** H@1=10.33% (from 1.41% zero-config, +633% relative) with a structural diagnosis: Freebase CVT mediator nodes deactivate the semantic attention term. This is the first explicit quantification of the CVT-adversarial property as the primary bottleneck for training-free systems on Freebase.

Every CEREBRUM answer is a fully traceable graph path. The system cannot hallucinate entities or relations absent from the KB. A domain expert can verify any conclusion against the source KB edge-by-edge without ML expertise.

Open problems. (1) *Discriminative re-ranking.* The H@10/H@1 gap is recoverable via a re-ranker using the CEREBRUM trace as a feature vector, preserving the training-free retrieval guarantee. (2) *Opaque-identifier KBs.* WebQSP’s CVT ceiling is addressable via KB preprocessing: mapping Freebase MIDs to readable entity labels restores the semantic attention signal. (3) *Automatic hop-count inference.* A training-free hop-count predictor (from question complexity or PathSchemaIndex schema length) would complete the zero-configuration guarantee.

Code, benchmark harnesses, and ParameterInitializer constant tables are available at <https://github.com/BrutalByte/CEREBRUM> under AGPL-3.0.

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